

Job description

SOC Analyst

The Key Digital Guardian in Modern Cybersecurity

Introduction

SOC Analysts are at the heart of cyber defense—monitoring, detecting, and responding to digital threats in real time.

What Is a SOC?

A **Security Operations Center (SOC)** acts as the cybersecurity control tower:

-  24/7 monitoring of networks and systems
 -  Continuous threat detection
 -  Rapid incident response and mitigation
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The SOC Analyst Role

SOC Analysts are the “digital first responders”:

-  Detect suspicious activity
 -  Trigger security responses
 -  Protect sensitive data and systems
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Tiered SOC Structure

SOC teams are divided into tiers based on expertise:

- **Tier 1** – First responders
- **Tier 2** – Incident analysts

- **Tier 3** – Threat hunters & strategists
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Tier 1 – The First Line

- 👁️ Monitors massive streams of alerts
 - 🧹 Filters out false positives
 - 📡 Escalates verified threats
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Tier 2 – Deep Investigation

- 🔍 Investigates incidents in depth
 - 🧩 Correlates logs and data
 - 📞 Coordinates the initial response
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Tier 3 – Senior Experts

- 🔧 Proactively hunts for hidden threats (Threat Hunting)
 - 🔬 Conducts digital forensics
 - 🛡️ Improves long-term security strategy
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A Day in the Life

- 📅 Review overnight alerts
 - 📡 Monitor dashboards and logs
 - 🔍 Analyze traffic and behavior
 - 📞 Respond to real-time threats
 - 📖 Document everything clearly
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Essential SOC Tools

SIEM (Security Information & Event Management)

What is it?

The central dashboard where SOC Analysts monitor and analyze security events.

What does it do?

- Collects data from all systems (firewalls, servers, applications, etc.)
- Correlates events to detect threats
- Triggers alerts when something suspicious occurs

Real-life example:

If a user tries to log in 10 times with the wrong password, the SIEM detects this and generates an alert.

Popular tools:

- Splunk
 - IBM QRadar
 - ArcSight
 - Microsoft Sentinel
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IDS/IPS: Intrusion Detection/Prevention Systems**What is it?**

Systems designed to detect (IDS) or block (IPS) network attacks.

What does it do?

- **IDS:** Detects and alerts on suspicious activity
- **IPS:** Can act automatically to block the attack

Real-life example:

If someone tries to exploit a vulnerability in your network, the IDS logs it, and the IPS can block the connection.

Popular tools:

- Snort (IDS)
 - Suricata
 - Cisco Firepower (IPS)
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EDR: Endpoint Detection and Response

What is it?

A security solution focused on protecting computers, laptops, and mobile devices (endpoints).

What does it do?

- Monitors every action on devices
- Detects suspicious behavior
- Allows incident investigation and response directly from the endpoint

Real-life example:

If a laptop downloads a malicious file and starts acting oddly, the EDR detects it and can automatically isolate it from the network.

Popular tools:

- CrowdStrike
 - SentinelOne
 - Microsoft Defender for Endpoint
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Technical Skills

- 🛠 Network and system administration
 - 🗝 Cybersecurity fundamentals
 - 📝 Familiarity with protocols, logs, and security tools
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Soft Skills

- 🧩 Critical thinking and pattern recognition
 - 🗣 Clear communication (technical and non-technical)
 - 🔥 Stress and time management
 - 🔍 Curiosity and willingness to learn continuously
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Virtual Network Lab – Simulating a Day in the Life of a SOC Analyst

Project Summary:

This lab simulates a typical workday for a SOC Analyst. It involves building and securing a corporate-style network, monitoring traffic, detecting intrusions, managing user access, and responding to potential threats—just like in a real-world SOC environment.

Simulated Daily Tasks (SOC Analyst Responsibilities):

Designing and Deploying a Secure Network

Built a virtual network using tools like Cisco Packet Tracer and GNS3, incorporating internal segments and a public-facing DMZ. Applied the CIA Triad (Confidentiality, Integrity, Availability) to guide security decisions.

Live Traffic Monitoring

Used Wireshark to inspect network packets for anomalies, identify DNS tunneling, suspicious HTTP traffic, and SMB-related issues—mirroring what a SOC analyst does when reviewing logs and alerts.

Configuring Firewall Rules

Implemented custom inbound/outbound rules with pfSense, including blocking unauthorized SSH, restricting web access, and setting NAT configurations for internal devices.

Intrusion Detection

Set up Snort IDS to detect threats in real time. Reviewed alert logs to identify scanning behavior, brute-force attempts, and policy violations.

User and Access Management

Configured Active Directory with user accounts, security groups, password policies, and access restrictions—essential for enforcing least privilege and managing insider risk.

Automation of Repetitive Tasks

Created PowerShell scripts to streamline account provisioning, apply security settings, and generate audit reports—similar to automation efforts in enterprise SOCs.

Key Skills Demonstrated:

- Threat detection and response
 - Firewall and access control configuration
 - Real-time traffic analysis and anomaly detection
 - Identity and Access Management (IAM)
 - Scripting and task automation
 - Network hardening and segmentation
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Tools & Technologies Used:

- Cisco Packet Tracer / GNS3 – Network simulation
- pfSense – Firewall and rule management
- Snort – Intrusion Detection System (IDS)
- Wireshark – Network packet analyzer
- Active Directory – IAM and policy enforcement
- PowerShell – Scripting and automation



Closing Thoughts

The SOC Analyst is a frontline cyber defender.
While technology evolves, the mission remains the same:
Keep the digital world secure.